

Mount Kenya University

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Unit Code: BIT3208

Unit Name: ADVANCED WEB DESIGN AND DEVELOPMENT

Lecturer Name: MR. MICHAEL NYORO

Semester/Academic Year: FOURTH YEAR

Project Title: TechHive Ecommerce

Technologies Used

- PHP (server-side scripting)
- MySQL (relational database)
- HTML5 and CSS3 (structure and styling)
- Bootstrap 5 (responsive UI framework)
- JavaScript (frontend validation and interactivity)
- XAMPP (local development environment)
- Figma (UI/UX design and prototyping)
- Git and GitHub (version control)

Github Link

https://github.com/immaculatemuli/techhive_ecommerce.git

Figma prototype

<https://www.figma.com/proto/kyLIU8WNEGKgExbtIBhopj/ecommerce?node-id=439-56&t=KH27XvfCzOfKu22w-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=383%3A1536>

Figma File

<https://www.figma.com/design/kyLIU8WNEGKgExbtIBhopj/ecommerce?node-id=383-1256&t=sOvq4IAF4Ru1EWtG-1>

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Week 1: Local Environment Setup

Title: Installation and Testing of Local Development Environment

Fig 1: Installation of XAMPP

(i had already installed XAMPP)

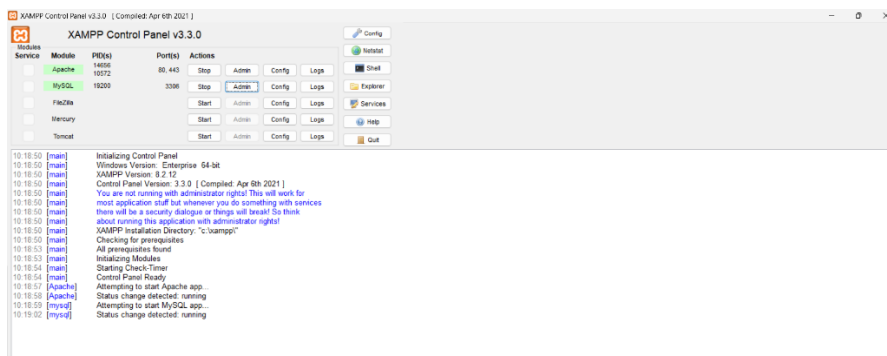


Fig 2: Apache and MySQL Running

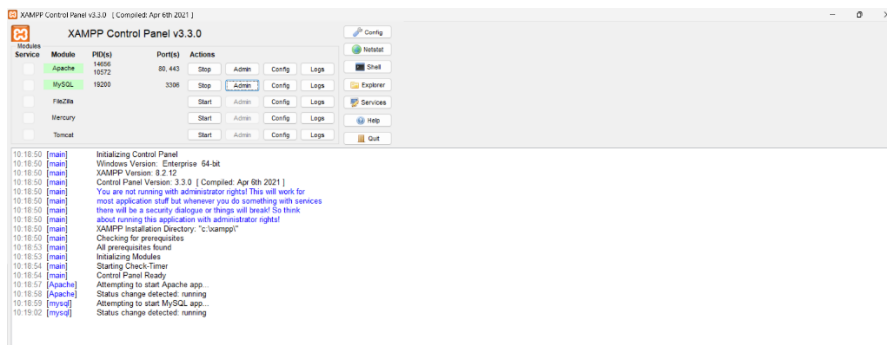


Fig 3: Localhost Test Page (HTML page)

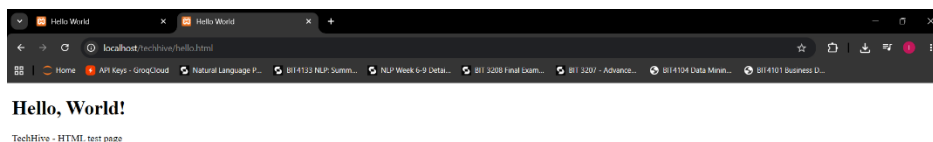


Fig 4: Hello World Test (PHP page)

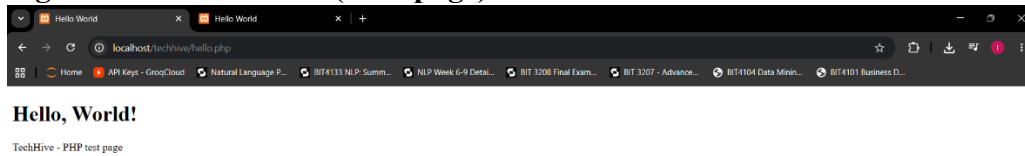
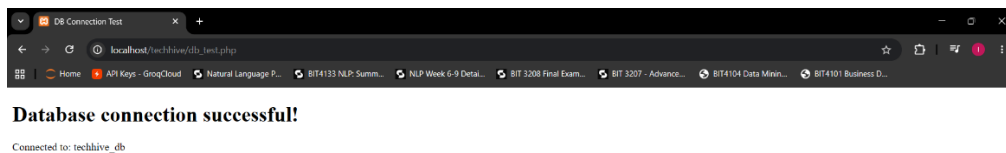


Fig 5: Database Connection Test



I set up my local development environment using XAMPP. Fig 1 shows the XAMPP control panel confirming successful installation on my machine. Fig 2 demonstrates Apache and MySQL services running, proving the server is active and ready to handle PHP requests. Fig 3 shows a localhost HTML test page, confirming Apache correctly serves static files through the browser. Fig 4 displays a PHP "Hello World" output, verifying that the PHP engine is properly configured and executing server-side code. Fig 5 shows a successful `mysqli_connect()` database connection, confirming PHP and MySQL are communicating correctly within the XAMPP environment.

Week 2: Wireframes and GUI Design

Title: User Interface Planning and System Design

Fig 1: Low fidelity Wireframes

LOGO



1.1 Sign in wireframe

LOGO



1.2 Sign up wireframe



1.3 Homepage wireframe



1.4 Dashbaord wireframe

Fig 2: Color Theme

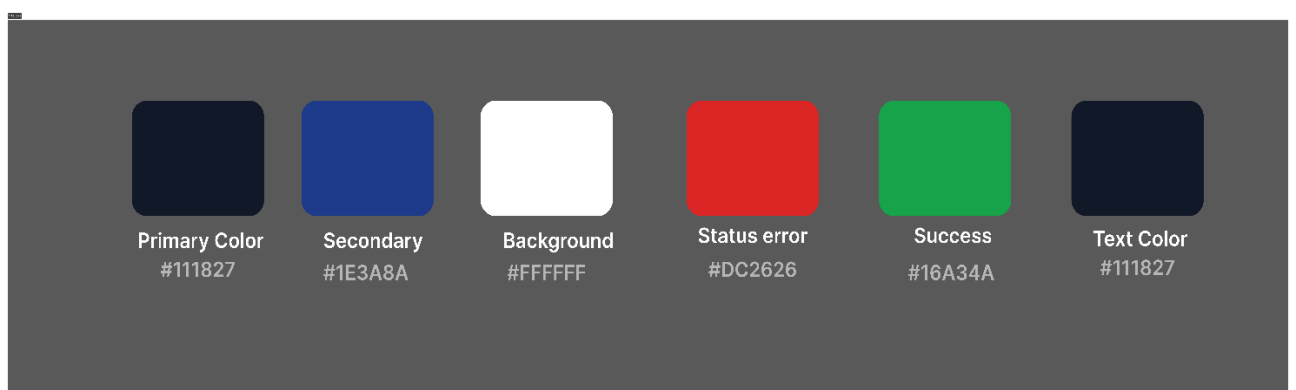
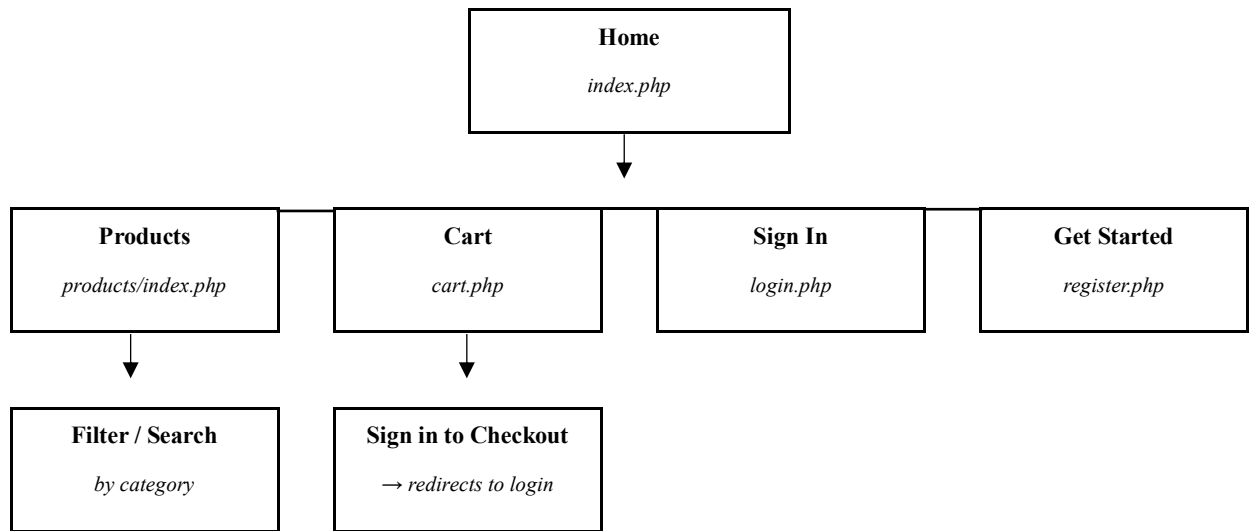
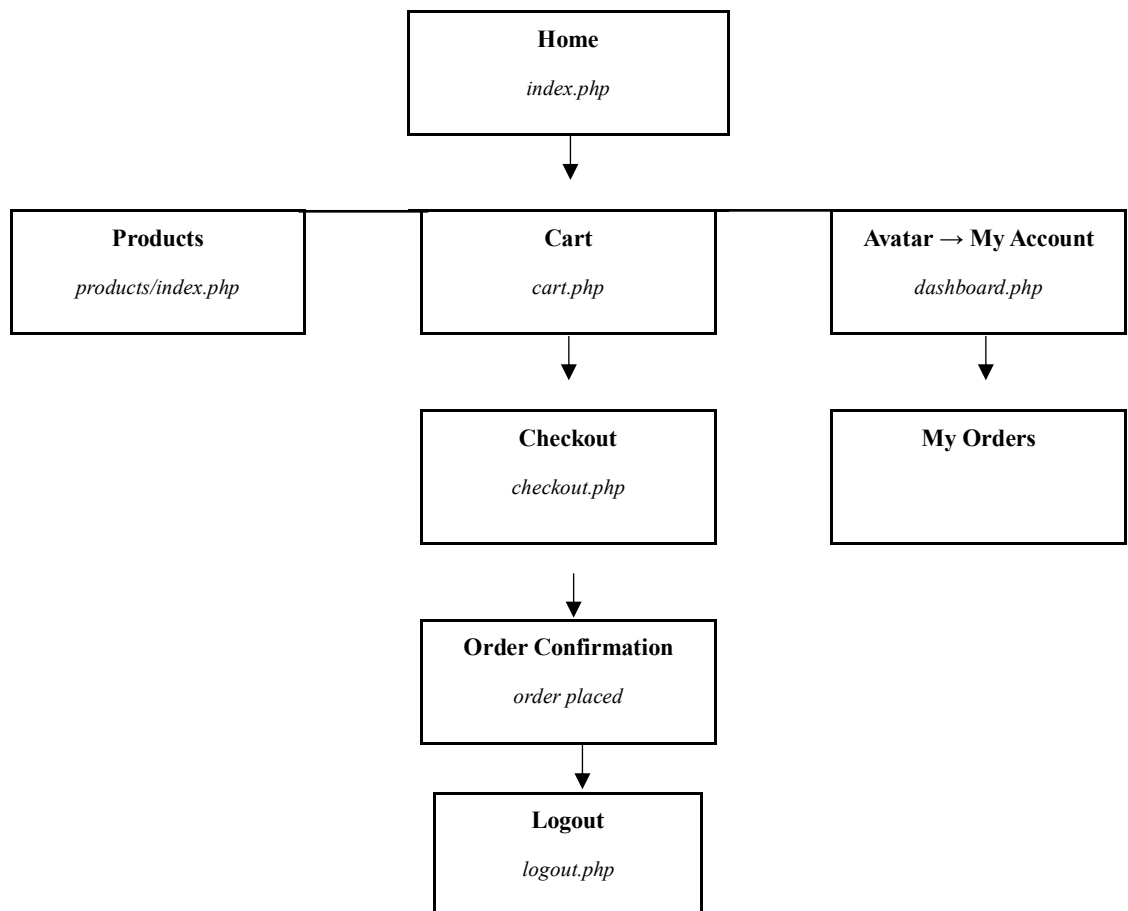


Fig 3: TechHive Navigation Structure

1. Public (Not Logged In)



2. Customer (Logged In)



3. Admin (Logged In as Admin)

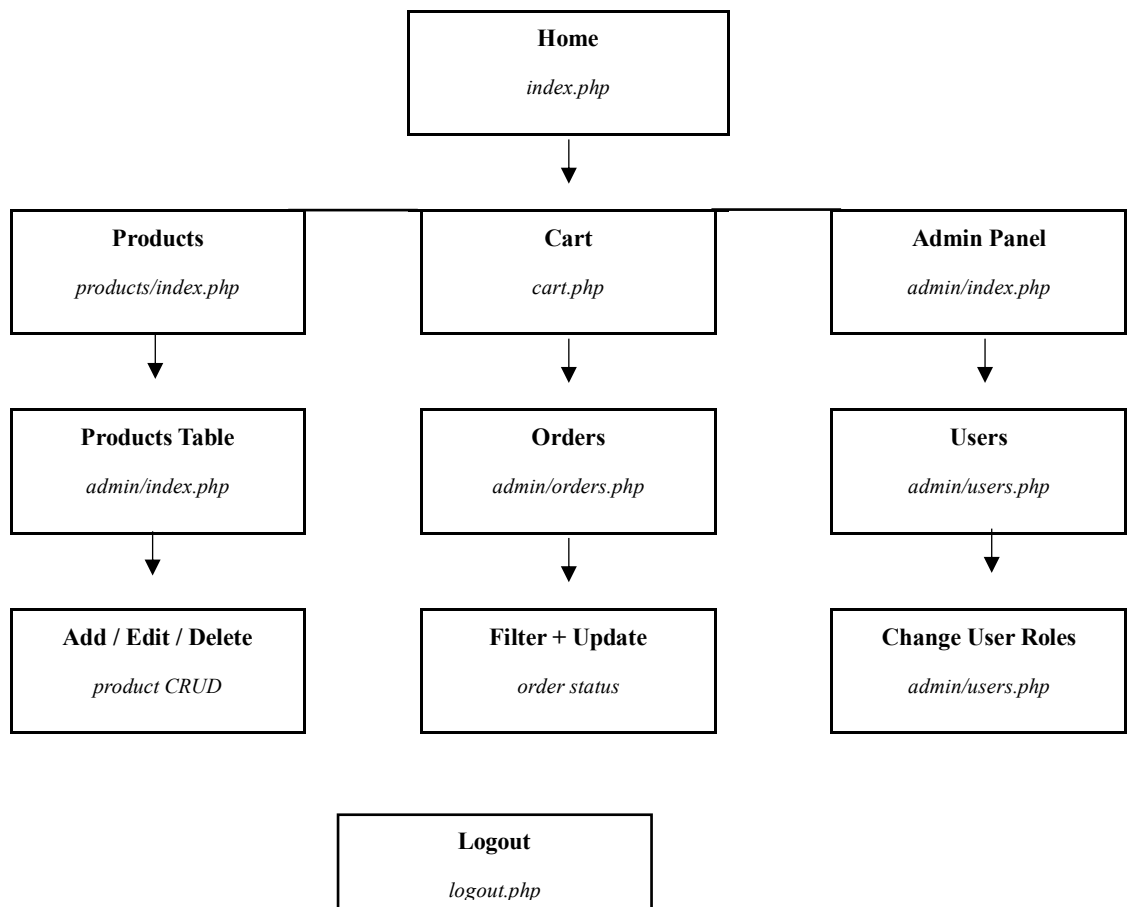


Fig 4: Mobile View Mockup

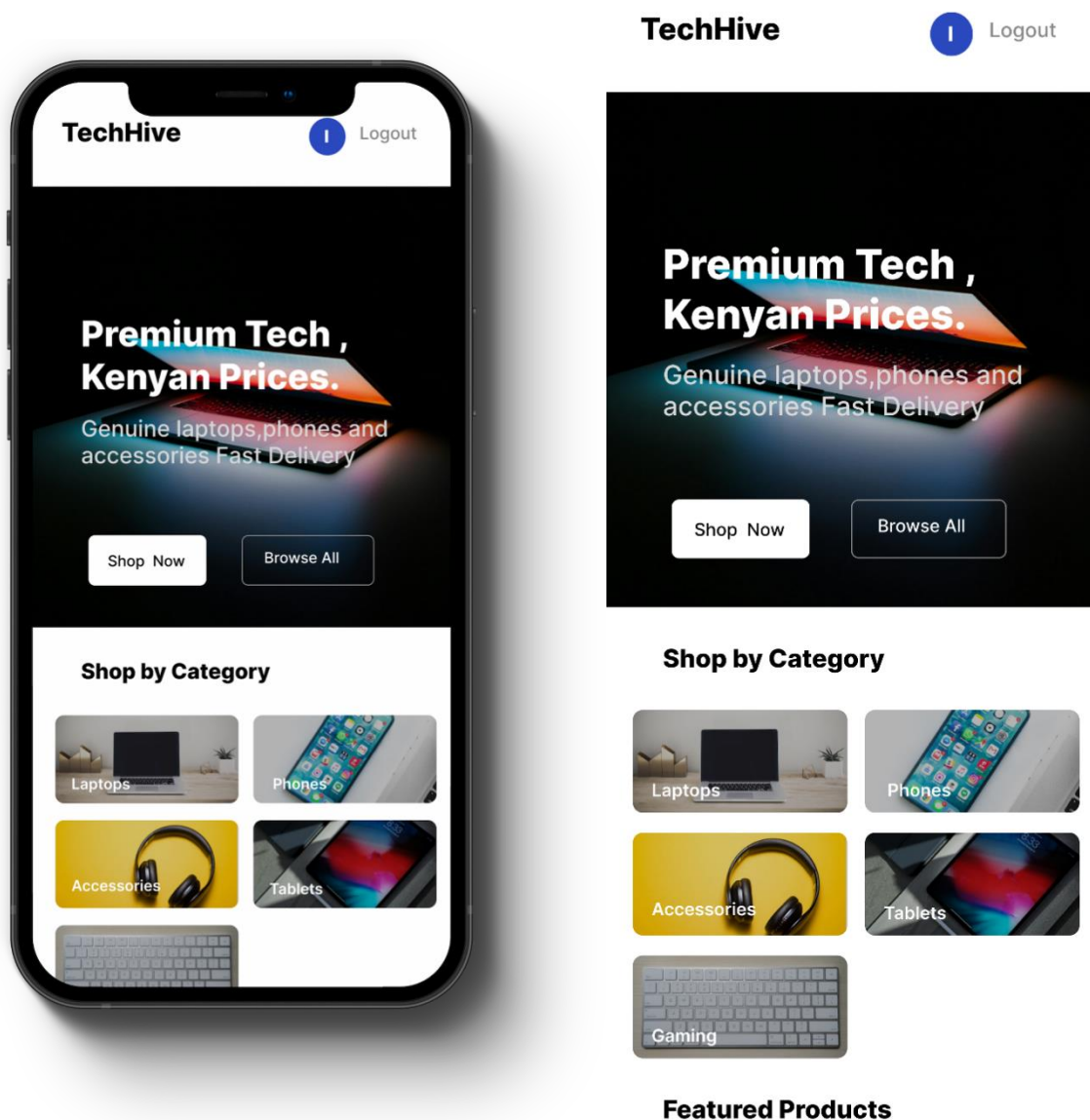


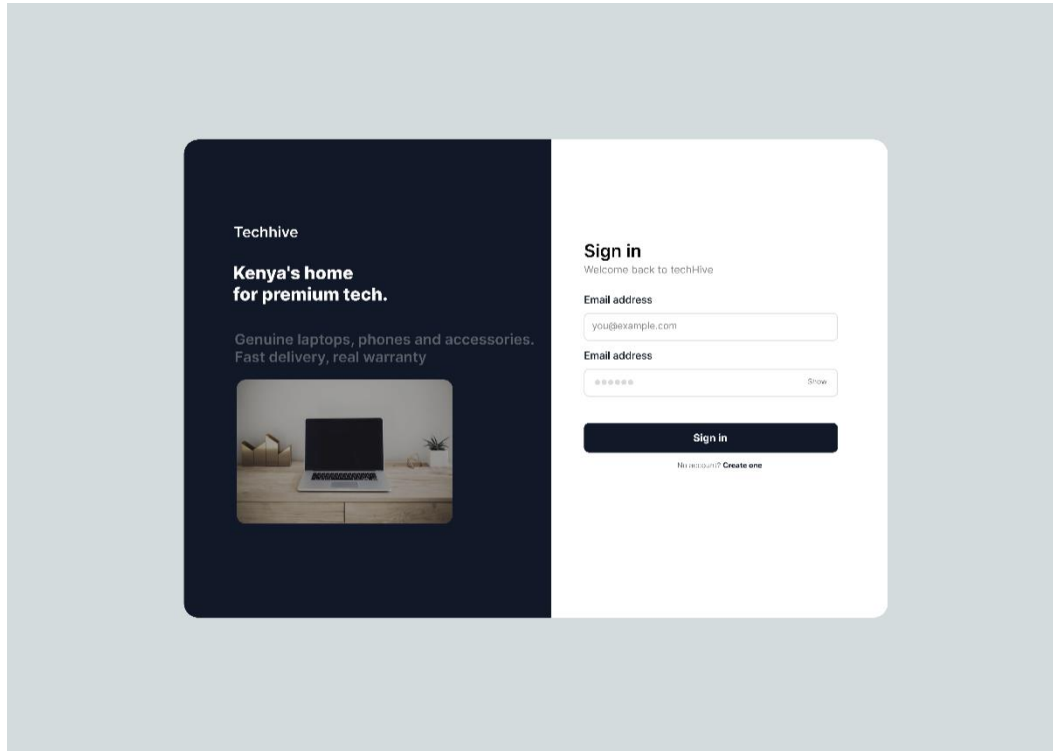
Fig 5: GUI Prototype in Figma

Figma prototype

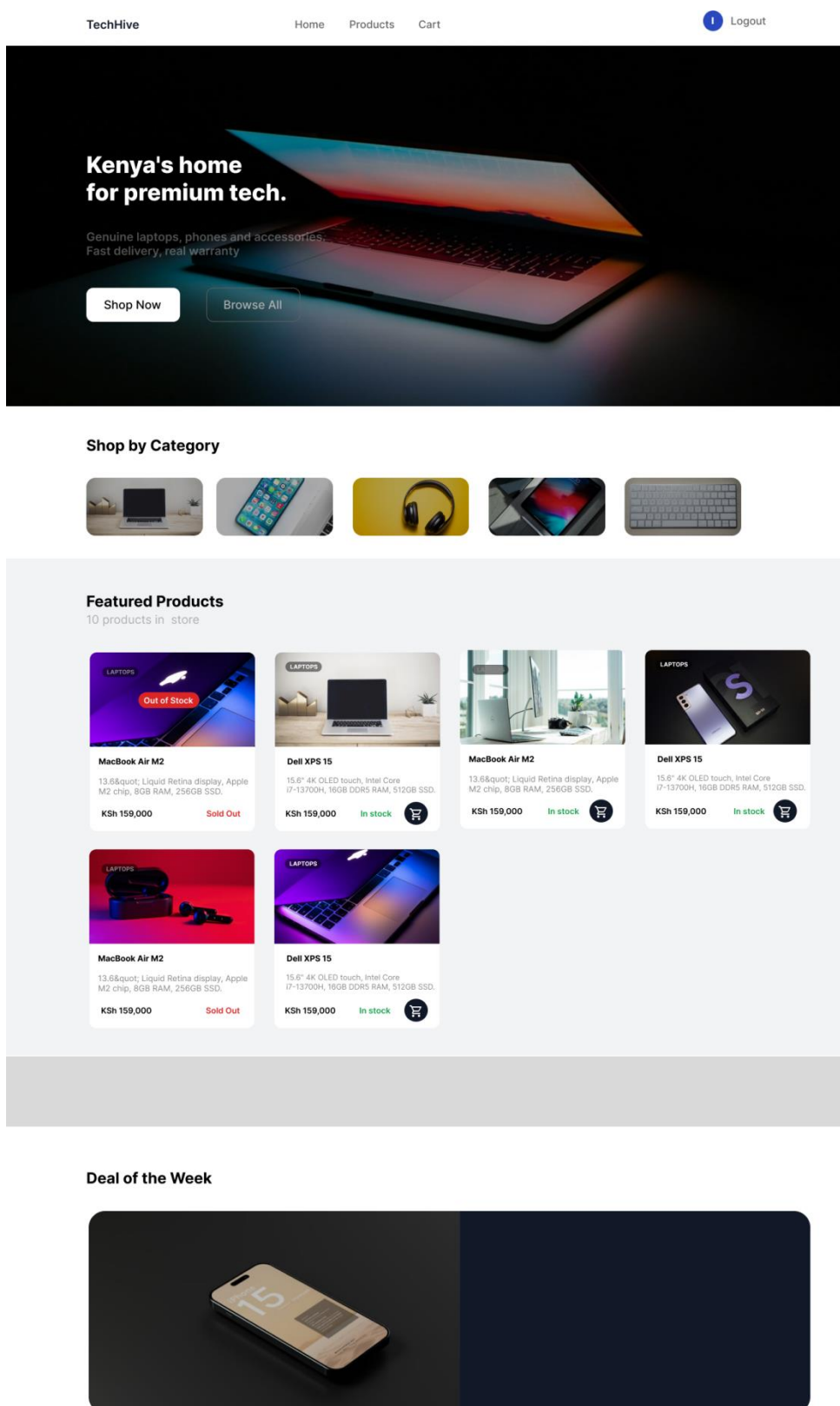
<https://www.figma.com/proto/kyLIU8WNEGKgExbtIBhopj/e-commerce?node-id=439-56&t=KH27XvfCzOfKu22w-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=383%3A1536>

Figma File

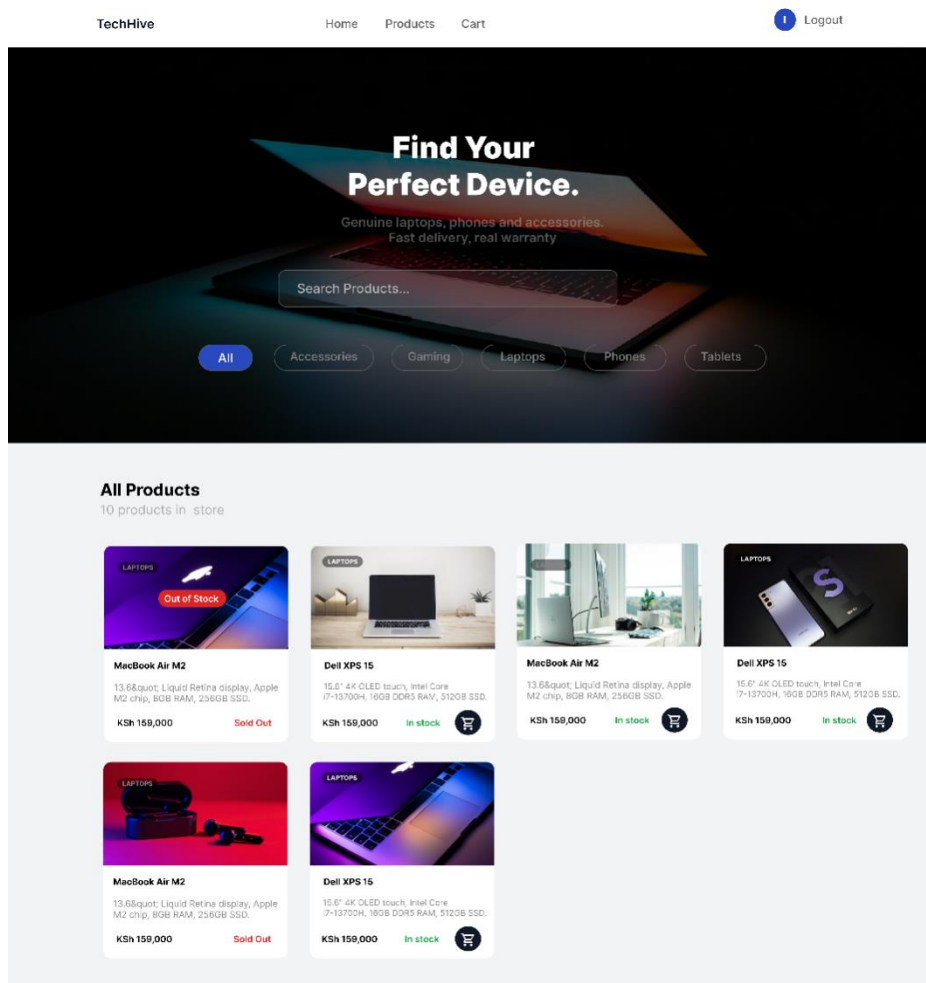
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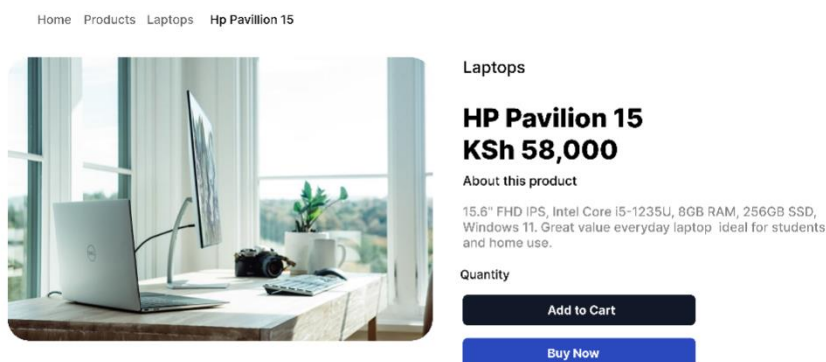
4.1: Sign in page



4.2: Homepage UI



4.3: Products page



4.4: Product details page

Fig 1 shows the low-fidelity wireframes designed in Figma covering the sign in page, sign up page, homepage and dashboard. Fig 2 shows the TechHive color theme built around a dark navy primary background to reflect a modern tech brand identity. Fig 3 shows the navigation structure diagram showing three user flows for Public, Customer and Admin navigation structure. Fig 4 shows the mobile view mockup confirming responsive layout considerations for small screens. The Figma prototype in Fig 5 applies the full color scheme and styling to the sign in page, homepage, products page and product details page showing the intended final user interface for TechHive.

Week 3: JavaScript and PHP Basics

Title: Frontend Interaction and Backend Foundations

Fig 1: JavaScript Form Validation

TechHive

Join thousands of tech shoppers.

Get exclusive deals, track your orders, and checkout faster with M-Pesa.

- ✓ Free delivery on orders above KSh 5,000
- ✓ 7-day hassle-free returns
- ✓ Secure M-Pesa & card payments

Create account

Username
lmm
Username is required.

Email address
lmm@gmail.com
Email is required.

Password
Min. 8 characters
Password is required.

Confirm password

Please confirm your password.

Create account

Already have an account? [Sign in](#)

Fig 2: Password Strength Checker

TechHive

Join thousands of tech shoppers.

Get exclusive deals, track your orders and checkout faster.

- ✓ Free delivery on orders above KSh 5,000
- ✓ 7-day hassle-free returns
- ✓ Secure M-Pesa & card payments

Create account

Username
lmm

Email address
client@gmail.com

Password

Too weak

- X At least 8 characters
- X One uppercase letter
- ✓ One lowercase letter
- X One number
- X One special character

Confirm password

Create account

Already have an account? [Sign in](#)

Create Account — TechHive

localhost/techhive/register.php

TechHive

Join thousands of tech shoppers.

Get exclusive deals, track your orders and checkout faster.

- ✓ Free delivery on orders above KSh 5,000
- ✓ 7-day hassle-free returns
- ✓ Secure M-Pesa & card payments

Create account

Username
lmm

Email address
client@gmail.com

Password
***** Show

- ✓ At least 8 characters
- ✓ One uppercase letter
- ✓ One lowercase letter
- ✓ One number
- ✓ One special character

Confirm password

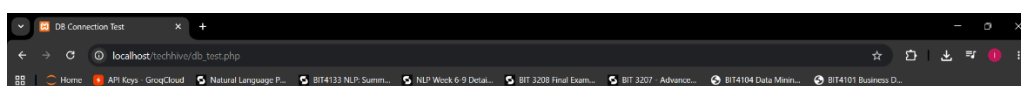
Create account

Already have an account? [Sign in](#)

Fig 3: PHP Syntax Practice

```
File Edit Selection View Go Run Terminal Help  
techhive  
db_test.php login.php M dynamic_input.php M register.php M index.php admin index.php \ M  
dynamic_input.php  
You, 16 hours ago | 1 author (You)  
1 <?php  
2 // PHP Syntax Practice - TechHive  
3 // Variables  
4 $storeName = "TechHive";  
5 $productName = "Gaming Laptop";  
6 $price = 85000;  
7 $inStock = true;  
8  
9 // Echo output  
10 echo "<h1>Welcome to " . $storeName . "</h1>";  
11 echo "<p>Product: " . $productName . "</p>";  
12 echo "<p>Price: KES " . $price . "</p>";  
13 echo "<p>In Stock: " . ($inStock ? "Yes" : "No") . "</p>";  
14 ?>
```

Fig 4: Database Connection Script



Database connection successful!

Connected to: techhive_db

Fig 5: Dynamic User Input Handling

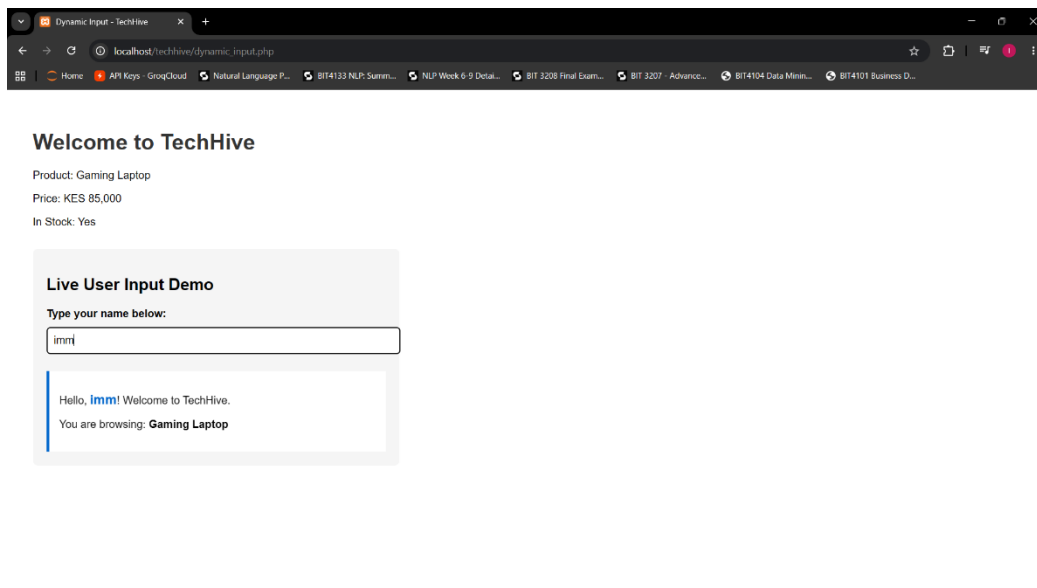


Fig 1: JavaScript was used on the frontend to validate form inputs on register.php, preventing empty or incorrectly formatted data from reaching the server. Fig 2: the password strength checker evaluates input in real time using keyboard events, giving users instant feedback by displaying weak or strong indicators before submission. Fig 3: Used PHP Syntax Practice demonstrating the use of PHP variables and echo output to dynamically display information generated on the server side. Fig 4 shows db_test.php returning a Connected Successfully message after mysqli_connect() established a connection to techhive_db, confirming PHP and MySQL are communicating correctly through XAMPP. Fig 5 shows dynamic_input.php where JavaScript DOM manipulation updates page content live as the user types without any page reload, demonstrating how frontend interactivity works independently of the server in TechHive.

Week 4: Authentication System

Title: Server-Side Components and Authentication System

Fig 1: Login form

The screenshot shows a web browser window with the URL `localhost/techhive/login.php`. The page features a dark blue sidebar on the left with the TechHive logo and a promotional message: "Kenya's home for premium tech. Genuine laptops, phones and accessories. Fast delivery, real warranty." Below this is an image of a laptop. The main content area is white and contains a "Sign in" form. The form includes a "Welcome back to TechHive" message, an "Email address" field with the placeholder `you@example.com`, a "Password" field with a "Show" toggle, a "Sign in" button, and a link to "Create one free" for users without an account.

Fig 2: Registration form

The screenshot shows a web browser window with the URL `localhost/techhive/register.php`. The page features a dark blue sidebar on the left with the TechHive logo and a promotional message: "Join thousands of tech shoppers. Get exclusive deals, track your orders and checkout faster." Below this are three bullet points: "Free delivery on orders above KSh 5,000", "7-day hassle-free returns", and "Secure M-Pesa & card payments". The main content area is white and contains a "Create account" form. The form includes a "Username" field with the placeholder `e.g. Imm`, an "Email address" field with the placeholder `Imm@gmail.com`, a "Password" field with a "Show" toggle and a list of requirements: "At least 8 characters", "One uppercase letter", "One lowercase letter", "One number", and "One special character". Below the password field is a "Confirm password" field with a "Show" toggle. A "Create account" button is at the bottom, followed by a link to "Sign in" for users who already have an account.

Fig 3: Simple Authentication System

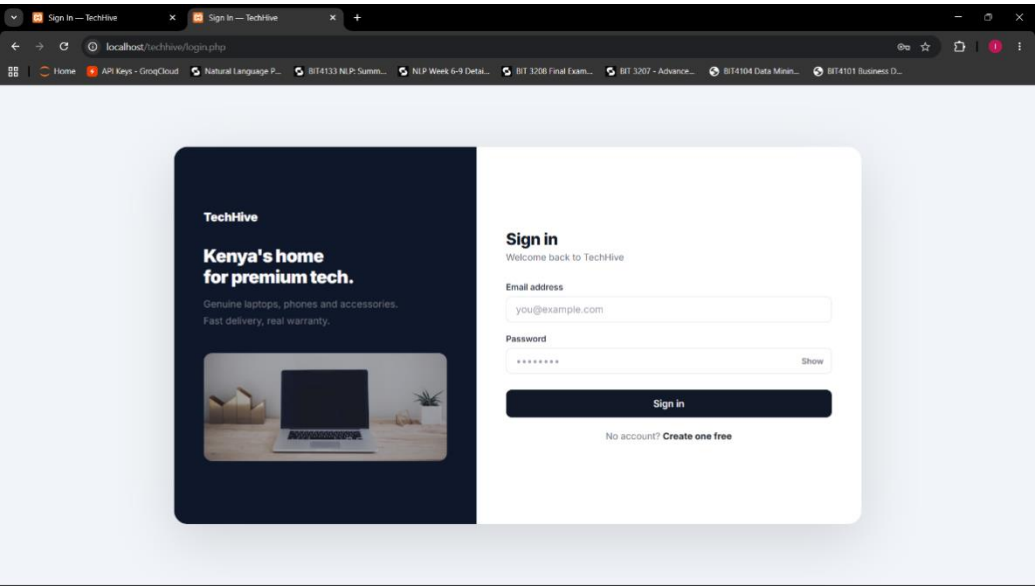
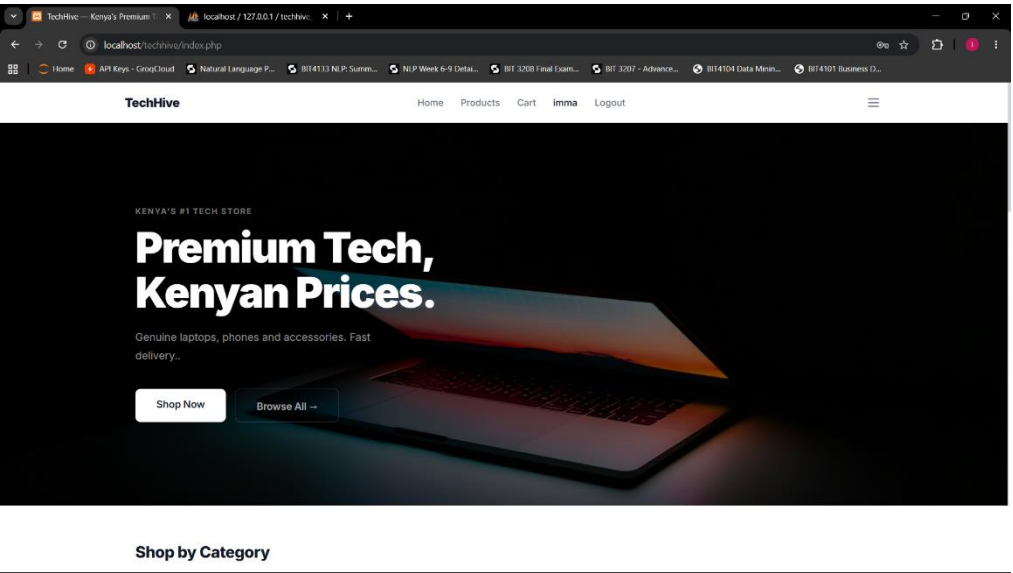
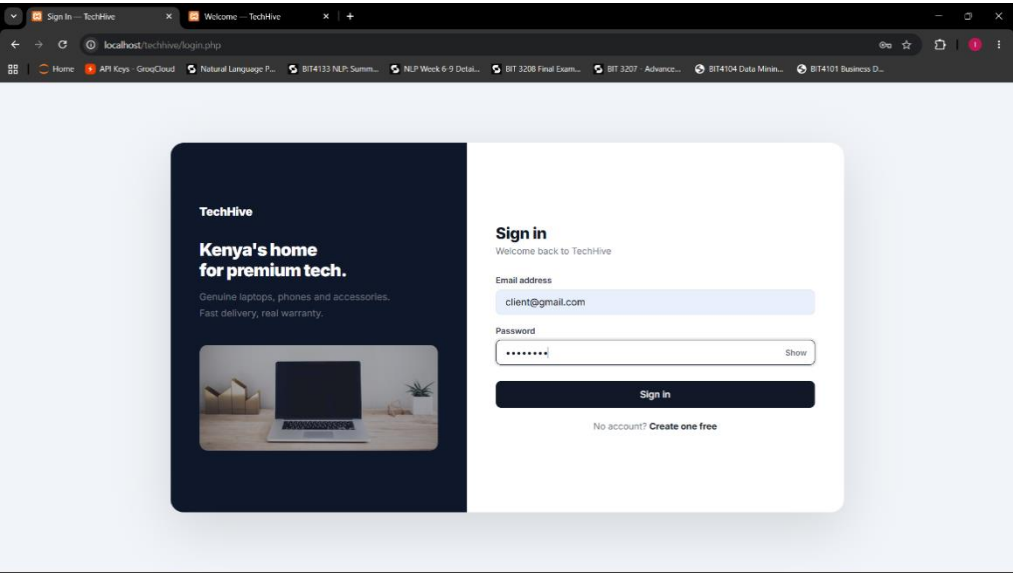
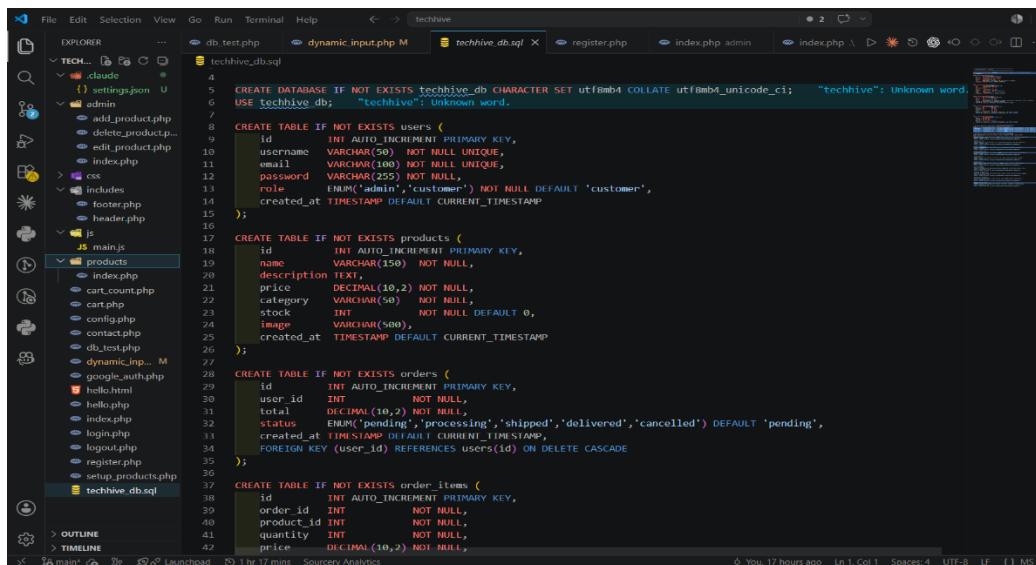


Fig 4: Backend Folder Structure



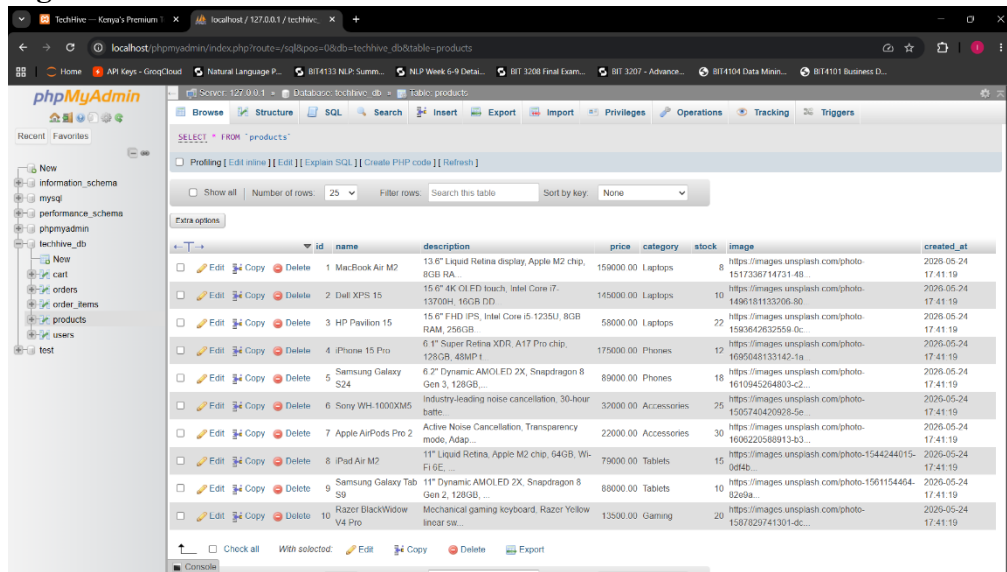
The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with folders like .cloud, .css, .includes, .js, .main.js, .products, .register.php, .index.php, .admin, .header.php, .footer.php, .dynamic_inp..., .google_auth..., .hello.html, .hello.php, .index.php, .login.php, .logout.php, .register.php, .setup_products.php, and .techhive_db.sql. The code editor shows the contents of techhive_db.sql, which contains SQL code to create a database and tables. The code is as follows:

```
4 CREATE DATABASE IF NOT EXISTS techhive_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
5 USE techhive_db;
6
7
8
9 CREATE TABLE IF NOT EXISTS users (
10   id INT AUTO_INCREMENT PRIMARY KEY,
11   username VARCHAR(50) NOT NULL UNIQUE,
12   email VARCHAR(100) NOT NULL UNIQUE,
13   password VARCHAR(255) NOT NULL,
14   role ENUM('admin','customer') NOT NULL DEFAULT 'customer',
15   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
16 );
17
18 CREATE TABLE IF NOT EXISTS products (
19   id INT AUTO_INCREMENT PRIMARY KEY,
20   name VARCHAR(150) NOT NULL,
21   description TEXT,
22   price DECIMAL(10,2) NOT NULL,
23   category VARCHAR(50) NOT NULL,
24   stock INT NOT NULL DEFAULT 0,
25   image VARCHAR(500),
26   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
27 );
28
29 CREATE TABLE IF NOT EXISTS orders (
30   id INT AUTO_INCREMENT PRIMARY KEY,
31   user_id INT NOT NULL,
32   total DECIMAL(10,2) NOT NULL,
33   status ENUM('pending','processing','shipped','delivered','cancelled') DEFAULT 'pending',
34   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
35   FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
36 );
37
38 CREATE TABLE IF NOT EXISTS order_items (
39   id INT AUTO_INCREMENT PRIMARY KEY,
40   order_id INT NOT NULL,
41   product_id INT NOT NULL,
42   quantity INT NOT NULL,
43   price DECIMAL(10,2) NOT NULL,
```

I learned server-side programming, PHP form integration, and session-based authentication. Fig 1 shows the login page where PHP reads `$_POST` data, queries the database, and uses `password_verify()` to authenticate the user. Fig 2 shows the registration page where form data is validated server-side using `filter_var()` and `preg_match()` before being stored with `password_hash()`. Fig 3 demonstrates the full authentication flow after a successful login, where `session_start()` and `$_SESSION` are used to store user session data, allowing the user to be redirected to the homepage after successful authentication and clicking logout triggers `session_destroy()` which clears the session and redirects back to login. Fig 4 shows the organised backend folder structure separating `admin/`, `includes/`, `css/`, and `js/` for maintainability.

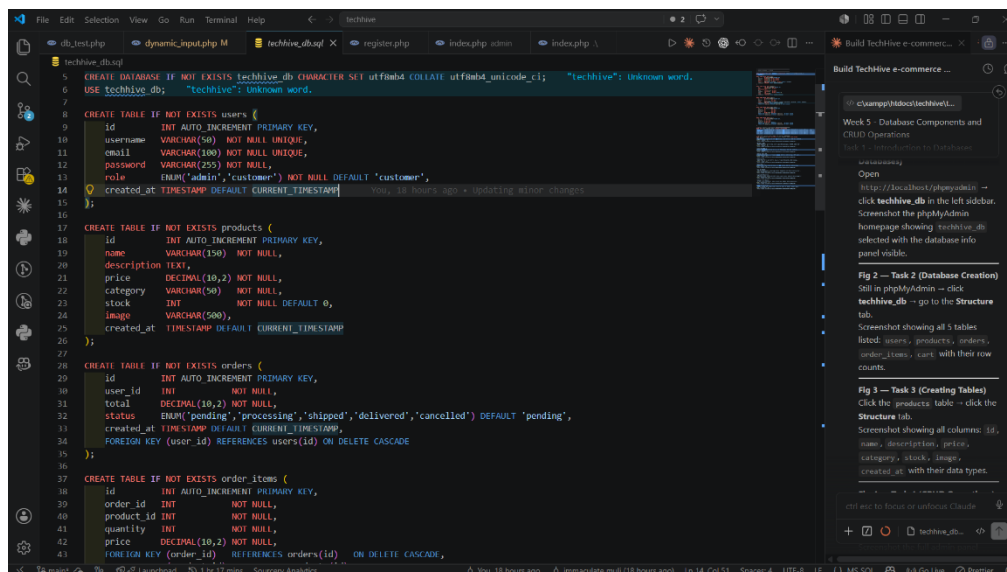
Week 5: Database Components and CRUD Operations

Fig 1: Database Creation



id	name	description	price	category	stock	image	created_at
1	MacBook Air M2	13.6" Liquid Retina display, Apple M2 chip, 8GB RAM	150000.00	Laptops	8	https://images.unsplash.com/photo-1517326714721-4b...	2026-05-24 17:41:19
2	Dell XPS 15	15.6" 4K OLED touch, Intel Core i7, 13700H, 16GB DDR5	145000.00	Laptops	10	https://images.unsplash.com/photo-1496181133206-80...	2026-05-24 17:41:19
3	HP Pavilion 15	15.6" FHD IPS, Intel Core i5-1235U, 8GB RAM, 256GB	58000.00	Laptops	22	https://images.unsplash.com/photo-1583642632559-0c...	2026-05-24 17:41:19
4	iPhone 15 Pro	6.1" Super Retina XDR, A17 Pro chip, 128GB, 48MP	175000.00	Phones	12	https://images.unsplash.com/photo-1685048133142-1a...	2026-05-24 17:41:19
5	Samsung Galaxy S24	6.2" Dynamic AMOLED 2X, Snapdragon 8 Gen 3, 128GB	89000.00	Phones	18	https://images.unsplash.com/photo-1610943264803-c2...	2026-05-24 17:41:19
6	Sony WH-1000XM5	Industry-leading noise cancellation, 30-hour batte...	32000.00	Accessories	25	https://images.unsplash.com/photo-1505740420928-5e...	2026-05-24 17:41:19
7	Apple AirPods Pro 2	Active Noise Cancellation, Transparency mode, Adap...	22000.00	Accessories	30	https://images.unsplash.com/photo-1606220588913-b3...	2026-05-24 17:41:19
8	iPad Air M2	11" Liquid Retina, Apple M2 chip, 64GB, Wi-Fi 6E	79000.00	Tablets	15	https://images.unsplash.com/photo-1544244015-9d4b...	2026-05-24 17:41:19
9	Samsung Galaxy Tab S9	11" Dynamic AMOLED 2X, Snapdragon 8 Gen 2, 128GB	86000.00	Tablets	10	https://images.unsplash.com/photo-1561154404-8269a...	2026-05-24 17:41:19
10	Razer BlackWidow V4 Pro	Mechanical gaming keyboard, Razer Yellow linear sw...	13500.00	Gaming	20	https://images.unsplash.com/photo-1587829741301-dc...	2026-05-24 17:41:19

Fig 2: Creating Tables



```
CREATE DATABASE IF NOT EXISTS techhive_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE techhive_db;

CREATE TABLE IF NOT EXISTS users (
  id INT AUTO INCREMENT PRIMARY KEY,
  username VARCHAR(50) NOT NULL UNIQUE,
  email VARCHAR(100) NOT NULL UNIQUE,
  password VARCHAR(255) NOT NULL,
  role ENUM('admin','customer') NOT NULL DEFAULT 'customer',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS products (
  id INT AUTO INCREMENT PRIMARY KEY,
  name VARCHAR(150) NOT NULL,
  description TEXT,
  price DECIMAL(10,2) NOT NULL,
  category VARCHAR(50) NOT NULL,
  stock INT NOT NULL DEFAULT 0,
  image VARCHAR(500),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS orders (
  id INT AUTO INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  total DECIMAL(10,2) NOT NULL,
  status ENUM('pending','processing','shipped','delivered','cancelled') DEFAULT 'pending',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);

CREATE TABLE IF NOT EXISTS order_items (
  id INT AUTO INCREMENT PRIMARY KEY,
  order_id INT NOT NULL,
  product_id INT NOT NULL,
  quantity INT NOT NULL,
  price DECIMAL(10,2) NOT NULL,
  FOREIGN KEY (order_id) REFERENCES orders(id) ON DELETE CASCADE,
```

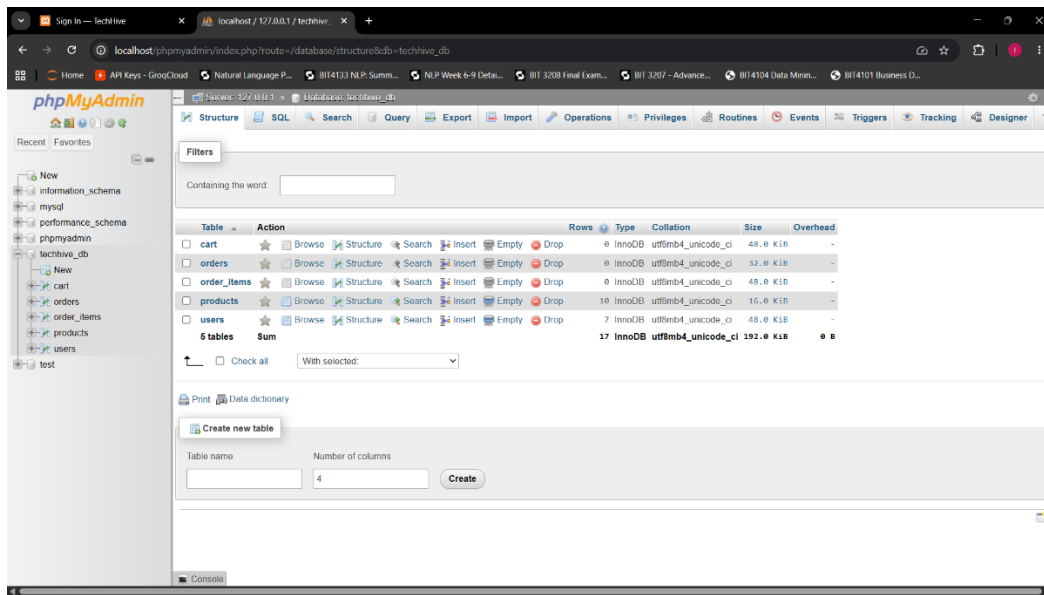


Fig 3: CRUD Operations

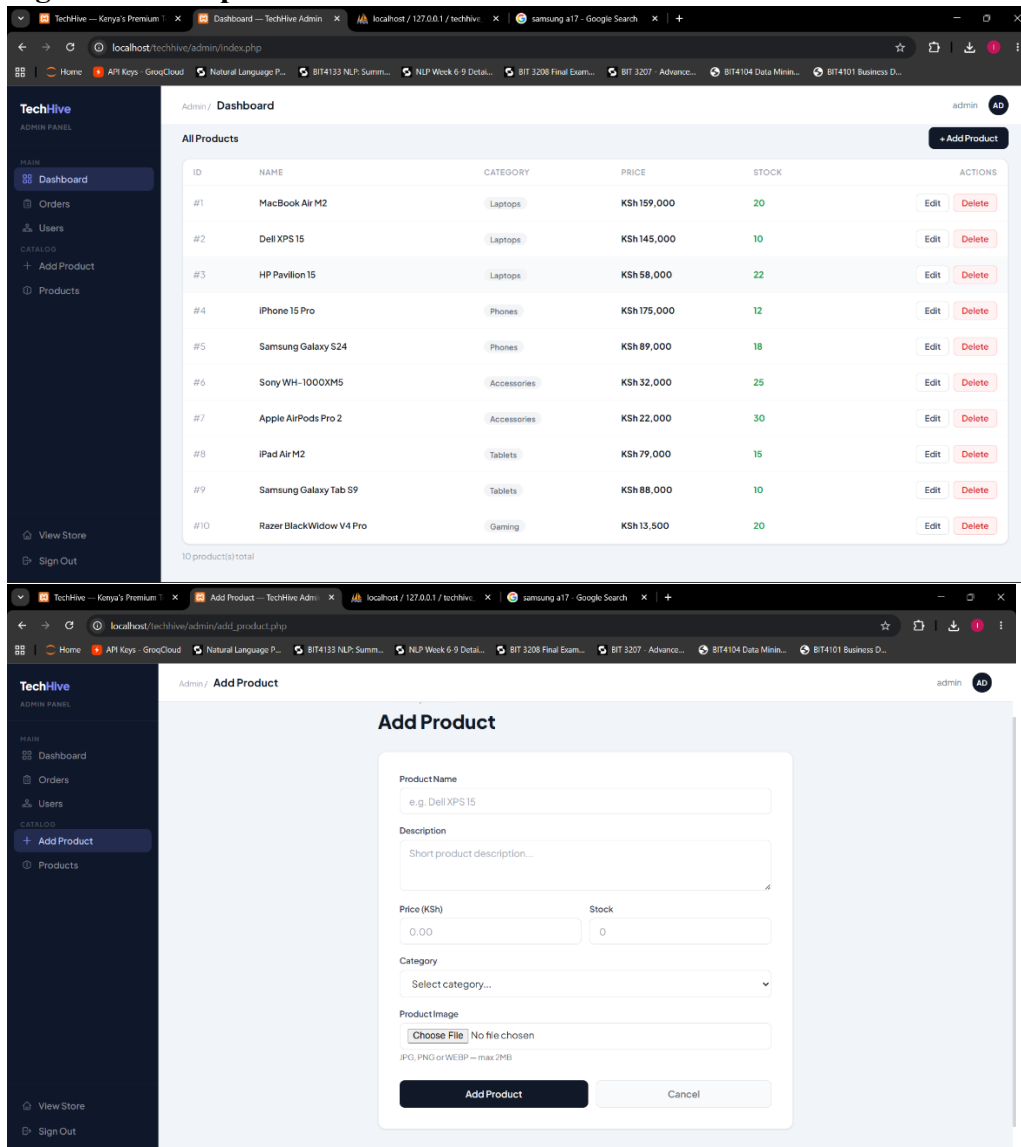
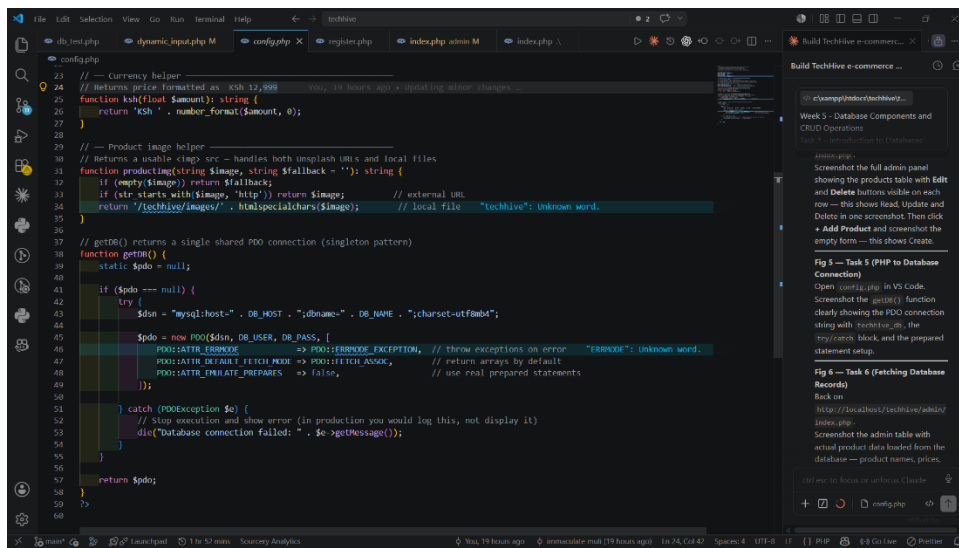


Fig 4: PHP to Database Connection



```
// config.php
23 // --- Currency helper
24 // Returns price formatted as KSh 12,999
25 function ksh(float $amount): string {
26     return "KSh " . number_format($amount, 0);
27 }
28
29 // --- Product image helper
30 // Returns a usable cimg src - handles both unplash URIs and local files
31 function producting(string $image, string $fallback = ""): string {
32     if (empty($image)) return $fallback;
33     if (str_starts_with($image, "http")) return $image; // external URL
34     return "techhive/images/" . htmlspecialchars($image); // local file "techhive": unknown word.
35 }
36
37 // getDB() returns a single shared PDO connection (singleton pattern)
38 function getDB() {
39     static $pdo = null;
40
41     if ($pdo === null) {
42         try {
43             $dsn = "mysql:host=" . DB_HOST . ";dbname=" . DB_NAME . ";charset=utf8mb4";
44             $pdo = new PDO($dsn, DB_USER, DB_PASS, [
45                 PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION, // throw exceptions on error "ERRMODE": unknown word.
46                 PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC, // return arrays by default
47                 PDO::ATTR_EMULATE_PREPARES => false, // use real prepared statements
48             ]);
49         } catch (PDOException $e) {
50             // Stop execution and show error (in production you would log this, not display it)
51             die("Database connection failed: " . $e->getMessage());
52         }
53     }
54
55     return $pdo;
56 }
57
58
59
60
61
62
63
64
65
66
67
68
```

Fig 5: Fetching Database Records

localhost - Keyga's Premium

Dashboard - Techhive Admin

localhost / 327.0.0.1 / techhive

samsung a17 - Google Search

+

+

localhost/phpmyadmin/index.php?route=/sql&pos=0&db=techhive_db&table=products

Home API Keys - GroqCloud Natural Language P... BI1413 NLP Summ... NLP Week 6-9 Data... BI1 3008 Final Exam... BI1 3207 - Advance... BI1404 Data Min... BI14101 Business D...

phpMyAdmin

Recent Favorites

New

information_schema

mysql

performance_schema

phpmyadmin

techhive_db

New

cart

orders

order_items

products

users

test

127.0.0.1 - Database: techhive_db Table: products

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

SELECT * FROM 'products'

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

		id	name	description	price	category	stock	image	created_at	
			1	MacBook Air M2	13.6" Liquid Retina display, Apple M2 chip, 8GB RAM...	150000.00	Laptops	8	https://images.unsplash.com/photo-1517338714731-48...	2026-05-24 17:41:19
			2	Dell XPS 15	15.6" 4K OLED touch, Intel Core i7-13700H, 16GB DDR...	145000.00	Laptops	10	https://images.unsplash.com/photo-149618133206-80...	2026-05-24 17:41:19
			3	HP Pavilion 15	15.6" FHD IPS, Intel Core i5-1235U, 8GB RAM, 256GB...	58000.00	Laptops	22	https://images.unsplash.com/photo-159364632559-dc...	2026-05-24 17:41:19
			4	iPhone 15 Pro	6.1" Super Retina XDR, A17 Pro chip, 128GB, 48MP Li...	175000.00	Phones	12	1695048133142-1a...	2026-05-24 17:41:19
			5	Samsung Galaxy S24	6.2" Dynamic AMOLED 2X, Snapdragon 8 Gen 3, 128GB...	89000.00	Phones	18	https://images.unsplash.com/photo-1610845064803-c2...	2026-05-24 17:41:19
			6	Sony WH-1000XM5	Industry-leading noise cancellation, 30-hour batte...	32000.00	Accessories	25	https://images.unsplash.com/photo-1505718420625-5e...	2026-05-24 17:41:19
			7	Apple AirPods Pro 2	Active Noise Cancellation, Transparency mode, Adap...	22000.00	Accessories	30	https://images.unsplash.com/photo-1906220588913-83...	2026-05-24 17:41:19
			8	iPad Air M2	11" Liquid Retina, Apple M2 chip, 64GB, Wi-Fi 6E, ...	79000.00	Tablets	15	https://images.unsplash.com/photo-1544244015-0d4b...	2026-05-24 17:41:19
			9	Samsung Galaxy Tab S9	11" Dynamic AMOLED 2X, Snapdragon 8 Gen 2, 128GB...	88000.00	Tablets	10	https://images.unsplash.com/photo-1561154464-8e9a...	2026-05-24 17:41:19
			10	Razer BlackWidow V4 Pro	Mechanical gaming keyboard, Razer Yellow linear sw...	13500.00	Gaming	20	https://images.unsplash.com/photo-1557829741301-dc...	2026-05-24 17:41:19

Check all

With selected

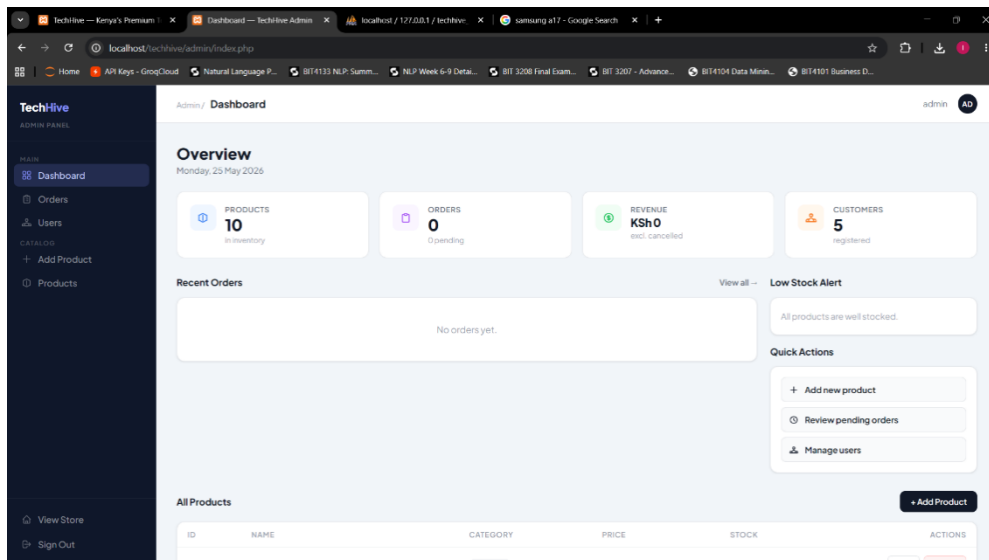
Edit

Copy

Delete

Export

Console



Week 5 introduced database-driven development for TechHive using MySQL through XAMPP. Fig 1 shows the techhive_db database created in phpMyAdmin containing all the required tables for the platform. Fig 2 shows the table structures created with proper data types, primary keys and foreign key relationships linking users, products, orders and cart tables together. Fig 3 demonstrates full CRUD operations working through the TechHive admin panel, allowing products to be created, read, updated and deleted dynamically from the database. Fig 4 shows the config.php file using PDO with a singleton getDB() function and prepared statements to prevent SQL injection across all database queries. Fig 5 shows products being fetched dynamically from techhive_db using SELECT queries and displayed in the admin panel, confirming successful PHP to database integration throughout TechHive.

Week 6: Database Integration and CRUD Operations

Fig 1: Database Creation

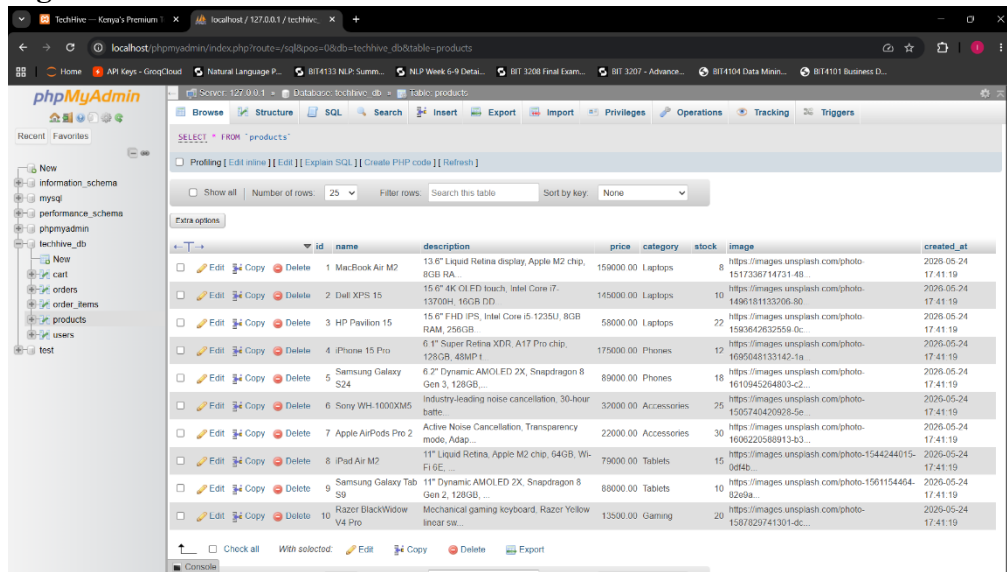


Fig 2: PHP to Database Connection

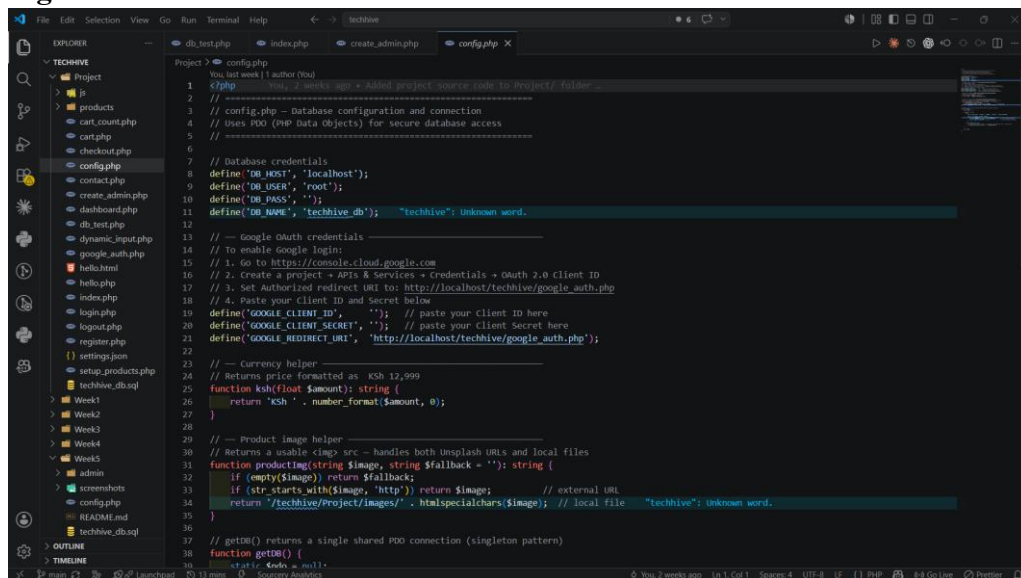


Fig 3: CREATE Operation

The screenshot shows the 'Add Product' form in the TechHive admin panel. The form is titled 'Add Product' and is located at the URL `localhost/techhive/Project/admin/add_product.php`. The form contains the following fields:

- ProductName:** MacBook Air M2
- Description:** grey MacBook Air M2 13inch screen size
- Price (KSh):** 80000
- Stock:** 19
- Category:** Laptops
- Product Image:** A file upload field with a 'Choose File' button and a 'download.jpg' file selected. Below the field, it says 'JPG, PNG or WEBP, max 2MB'.

At the bottom of the form, there are two buttons: 'Add Product' and 'Cancel'.

Fig 4: READ Operation

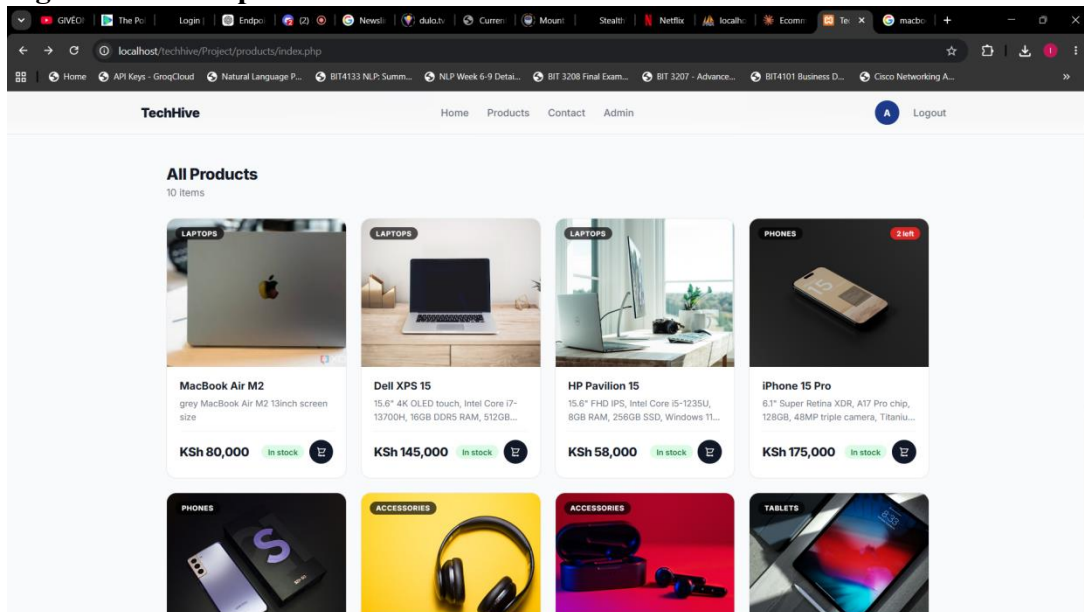


Fig 5: UPDATE and DELETE Operations

The screenshot shows the 'Edit Product' form in the TechHive Admin Panel. The form is for product ID #13, 'MacBook Air M2'. The fields are as follows:

Field	Value
Product Name	MacBook Air M2
Description	grey MacBook Air M2 13inch screen size
Price (KSh)	80000.00
Stock	19
Category	Laptops

Below the form, there is a 'Current Image' section showing a small image of the MacBook Air M2. Below that is a 'Choose File' button and a note: 'No file chosen. Leave blank to keep current image'.

The screenshot shows the 'All Products' table in the TechHive Admin Panel. A confirmation dialog is open, asking to delete product #13, 'MacBook Air M2'. The table contains the following data:

ID	NAME	CATEGORY	PRICE	STOCK	ACTIONS
#13	MacBook Air M2	Laptops	KSh 80,000	19 in stock	Edit Delete
#2	Dell XPS 15	Laptops	KSh 145,000	10 in stock	Edit Delete
#3	HP Pavilion 15	Laptops	KSh 58,000	22 in stock	Edit Delete
#4	iPhone 15 Pro	Phones	KSh 175,000	2 left	Edit Delete
#5	Samsung Galaxy S24	Phones	KSh 89,000	18 in stock	Edit Delete
#6	Sony WH-1000XM5	Accessories	KSh 32,000	25 in stock	Edit Delete
#7	Apple AirPods Pro 2	Accessories	KSh 22,000	30 in stock	Edit Delete
#8	iPad Air M2	Tablets	KSh 79,000	15 in stock	Edit Delete
#9	Samsung Galaxy Tab S9	Tablets	KSh 88,000	10 in stock	Edit Delete
#10	Razer BlackWidow V4 Pro	Gaming	KSh 13,500	20 in stock	Edit Delete

At the bottom of the table, it says '10 product(s) total'.

Week 6 focused on integrating TechHive with MySQL through full CRUD operations. Fig 1 shows techhive_db created in phpMyAdmin containing all required tables for the platform. Fig 2 shows config.php using PDO with a singleton getDB() function and prepared statements preventing SQL injection. Fig 3 demonstrates the CREATE operation through the admin add product form inserting records using INSERT INTO. Fig 4 shows the READ operation fetching all products dynamically using SELECT and displaying them on the homepage and admin panel. Fig 5 demonstrates UPDATE modifying product details and DELETE removing records from techhive_db permanently.

Week 7: User Authentication and Session Management

Fig 1: User Registration with Password Hashing

The image displays two screenshots from a web application. The top screenshot shows a 'Create account' form for 'TechHive'. The form includes fields for Username, Email address, Password, and Confirm password. The Password field has a 'Show' toggle and a list of requirements: At least 8 characters, One uppercase letter, One lowercase letter, One number, and One special character. The bottom screenshot shows the phpMyAdmin interface for the 'techhive_db' database, displaying the 'users' table. The table contains 6 rows of user data, including usernames, emails, roles, and hashed passwords.

Create account

Username
immaculate

Email address
mulimaculate@gmail.com

Password
***** Show

- ✓ At least 8 characters
- ✓ One uppercase letter
- ✓ One lowercase letter
- ✓ One number
- ✓ One special character

Confirm password

Create account

Already have an account? [Sign in](#)

phpMyAdmin

Recent Favorites

techhive_db

- New
- information_schema
- mysql
- performance_schema
- phpmyadmin
- techhive_db
 - New
 - cart
 - orders
 - order_items
 - products
 - users
 - test

SELECT * FROM users ORDER BY password ASC

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Restore column order Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	id	username	email	role	password	created_at
☐	2	superadmin	super@techhive.com	customer	\$2y\$10\$92IXUNpkjO0rQ5byMl:Ye4oKoEa3Ro9lCj/og/at2...	2026-05-24 12:04:48
☐	3	john_doe	john@example.com	customer	\$2y\$10\$92IXUNpkjO0rQ5byMl:Ye4oKoEa3Ro9lCj/og/at2...	2026-05-24 12:04:48
☐	4	jane_smith	jane@example.com	customer	\$2y\$10\$92IXUNpkjO0rQ5byMl:Ye4oKoEa3Ro9lCj/og/at2...	2026-05-24 12:04:48
☐	5	mike_jones	mike@example.com	customer	\$2y\$10\$92IXUNpkjO0rQ5byMl:Ye4oKoEa3Ro9lCj/og/at2...	2026-05-24 12:04:48
☐	15	Admin	admin@techhive.com	admin	\$2y\$10\$96H-NDekHauJcVgDyKebHYvENnK2dfsZuuc5q...	2026-05-27 10:12:10
☐	14	imma	client@gmail.com	customer	\$2y\$10\$K5/QB4sH790IGB1M8slUuO3JbJEZRP/6hmMOVcTa5...	2026-05-25 12:02:42
☐	6	immaculate	mulimaculate@gmail.com	customer	\$2y\$10\$NdxEYphqBQWovEozmRpsbe5Tp3ZRbjV5MGKn3QQpc...	2026-05-24 12:32:19

Check all With selected Edit Copy Delete Export

Show all Restore column order Number of rows: 25 Filter rows: Search this table Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Console

Fig 2: User login

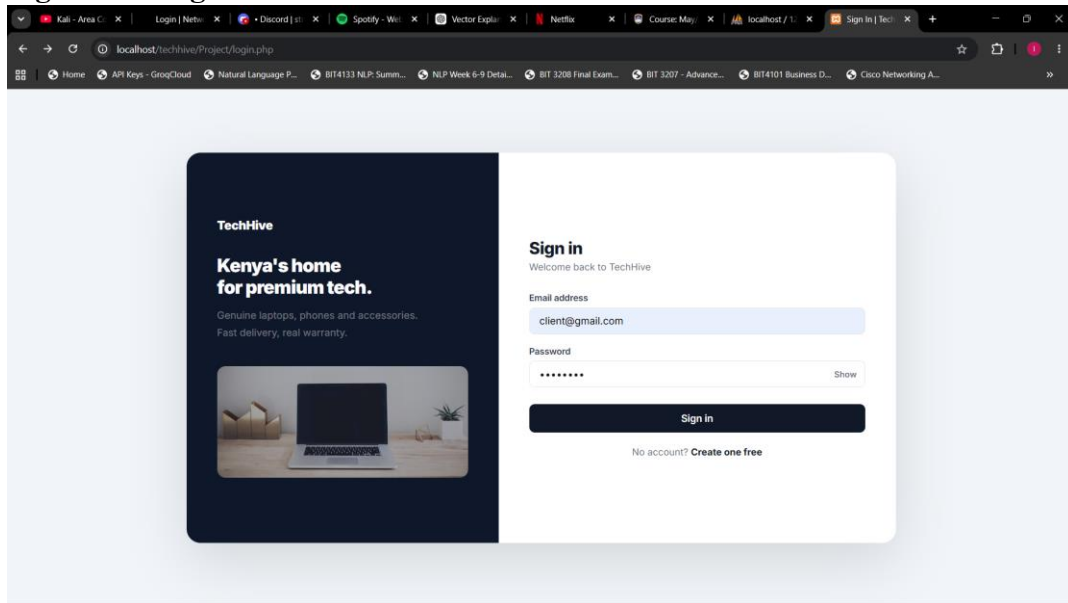


Fig 3: Session-Based Successful

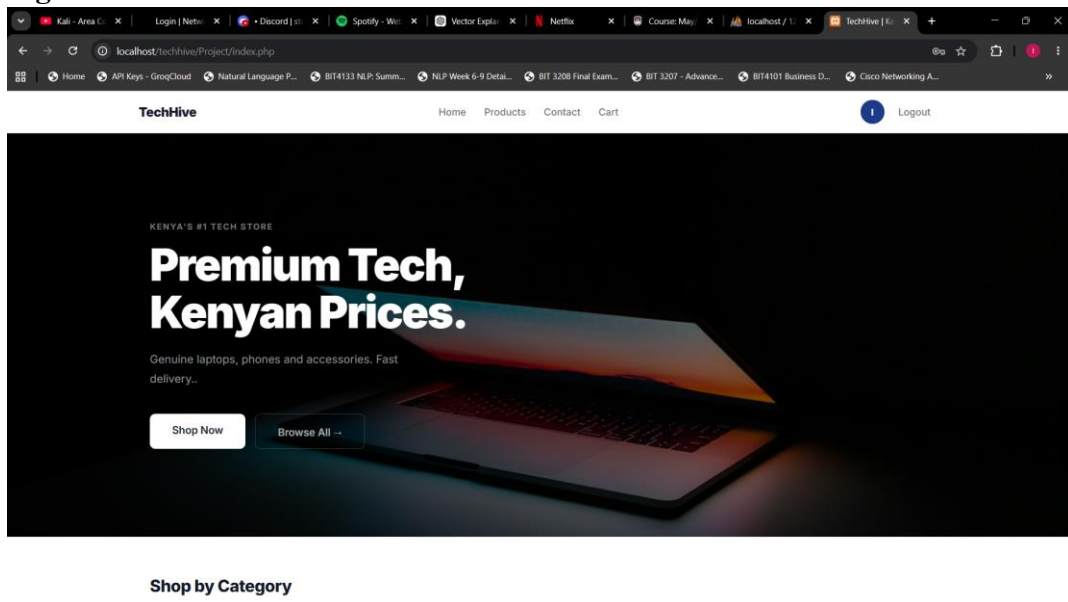
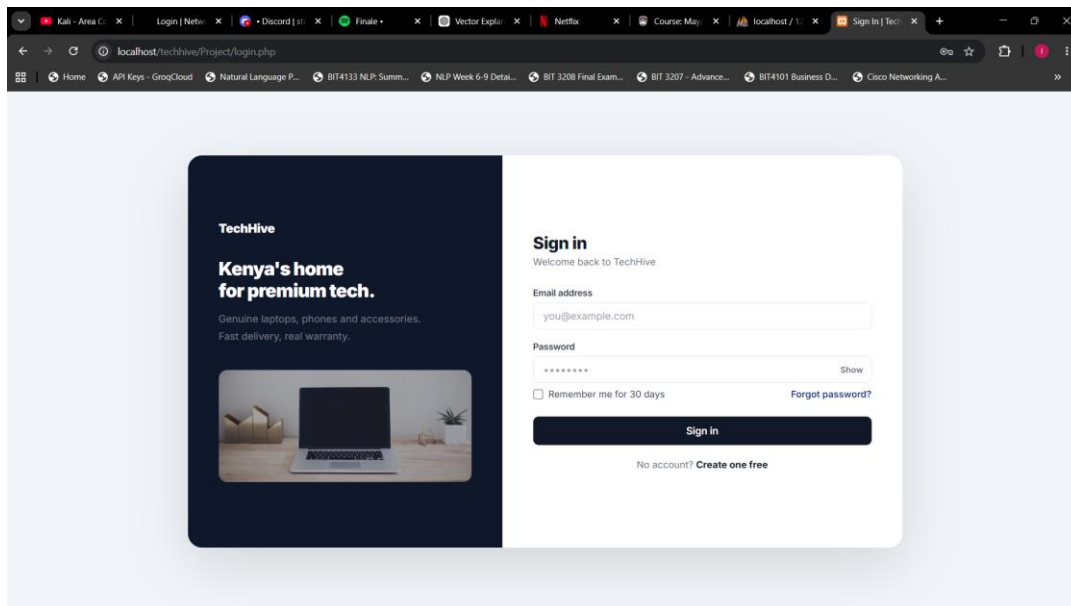
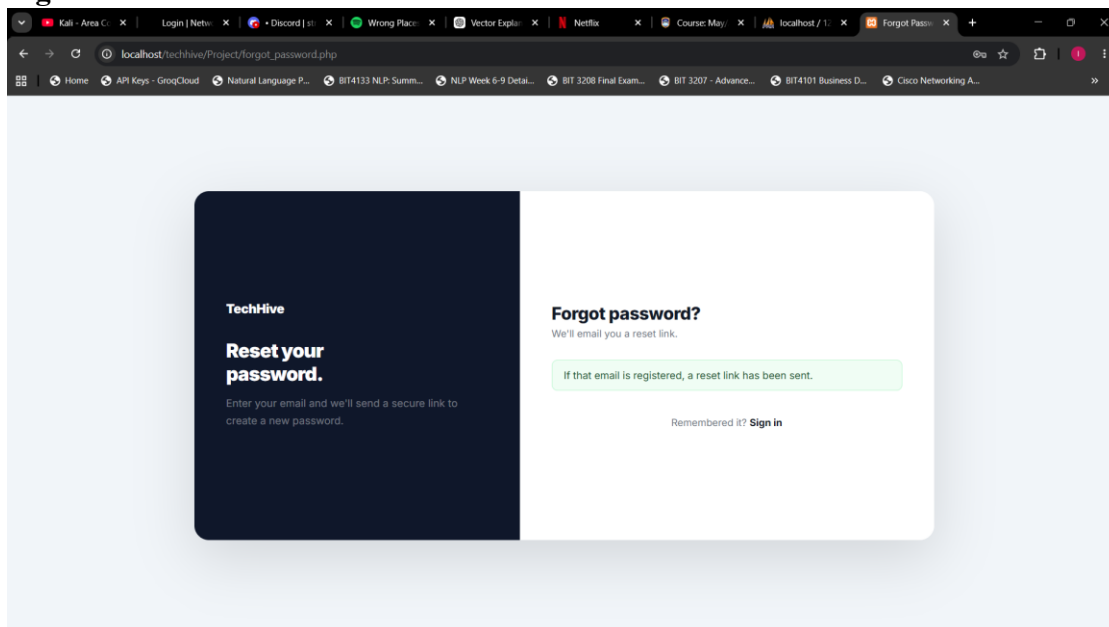


Fig 4: Logout Functionality



BONUS FEATURES

Fig 1: Password reset



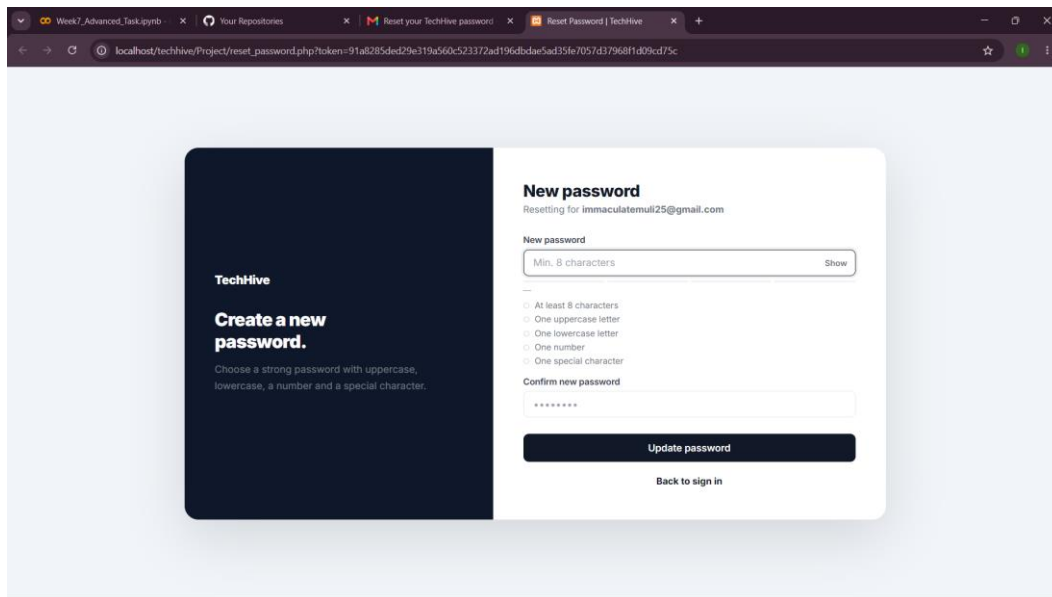


Fig 2: Email Verification

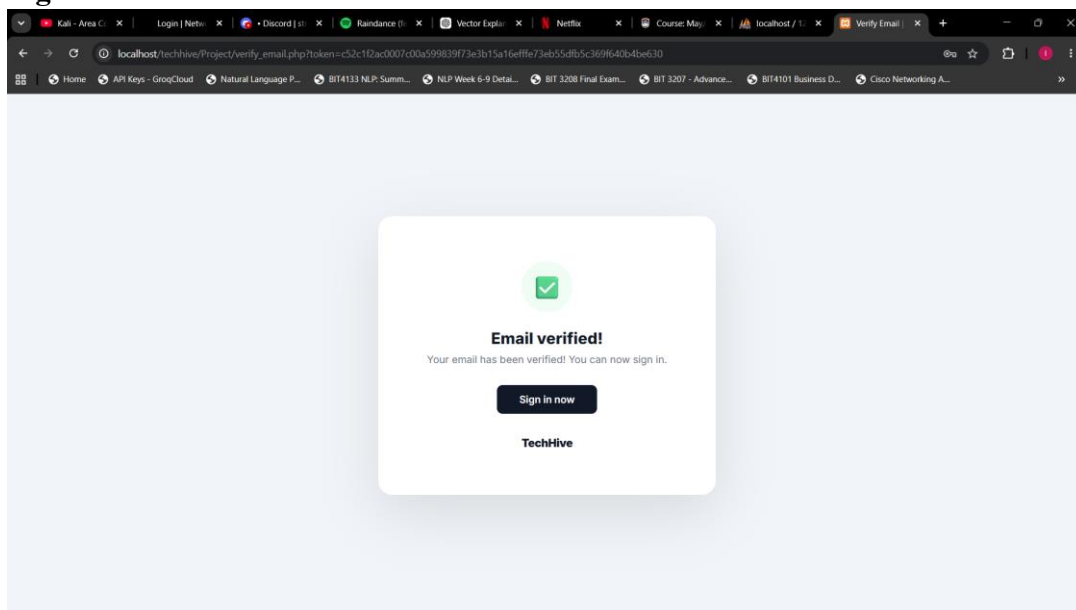


Fig 3: Two-Factor Authentication (2FA)

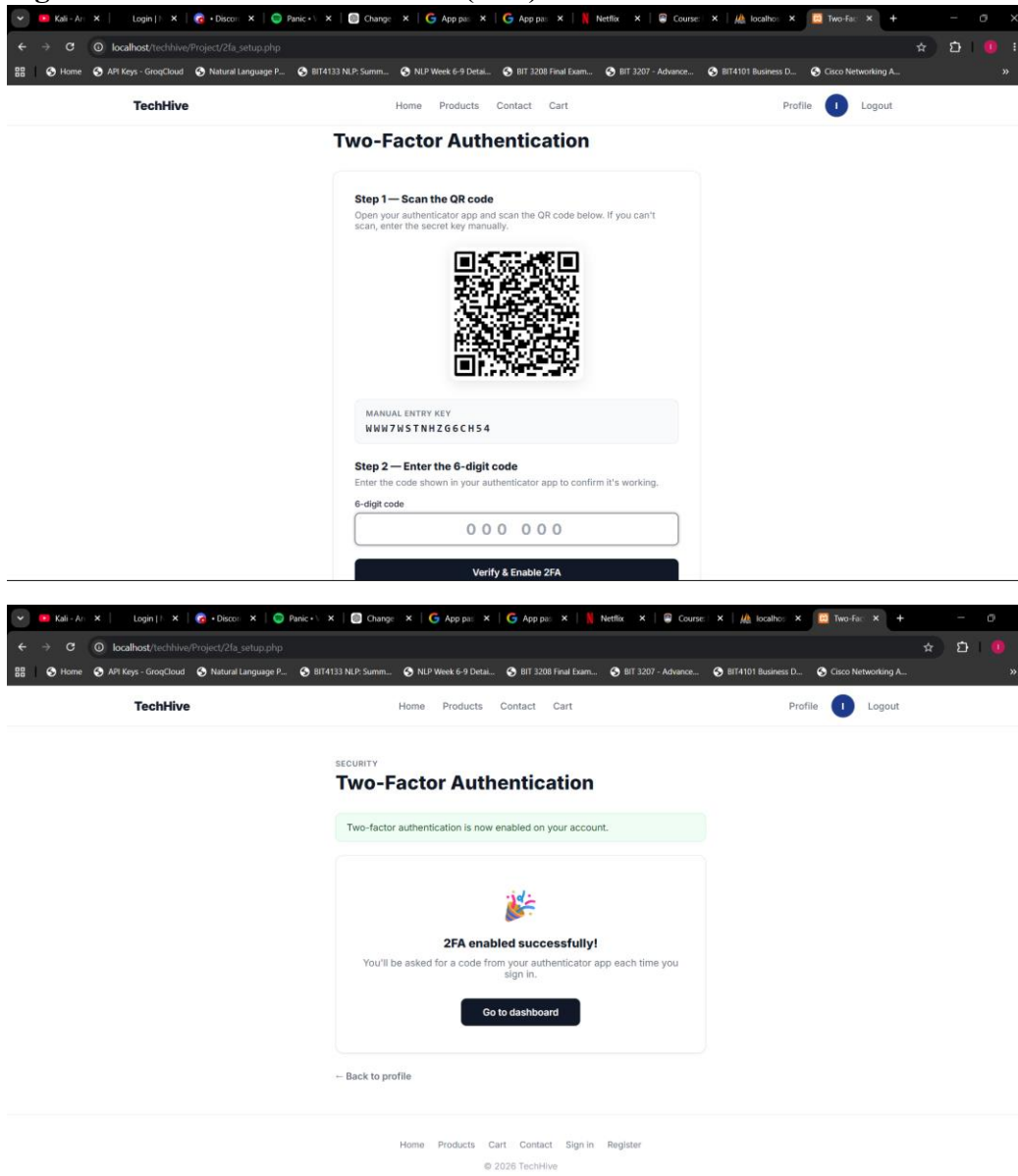


Fig 4: User Profile Management

The image displays two screenshots of the TechHive user profile management interface. The top screenshot shows the 'Profile Info' tab with fields for Username, Email address, Phone number, and Bio. The bottom screenshot shows the 'My Profile' page with a success message 'Profile updated successfully.' and the same form fields.

Top Screenshot: Profile Info

Username: Immaculate muli
Email address: immaculatemuli25@gmail.com
Phone number (optional): +254 7XX XXX XXX
Bio (optional): hey there

Bottom Screenshot: My Profile

Profile updated successfully.

Username: Immaculate muli
Email address: immaculatemuli25@gmail.com
Phone number (optional): +254 7XX XXX XXX
Bio (optional): hey there

Week 7 built a complete authentication system for TechHive. Fig 1 shows register.php hashing passwords using password_hash() before storing them in techhive_db. Fig 2 shows login.php verifying credentials using password_verify() and creating a session on success. Fig 3 shows the homepage displaying the logged-in username from \$_SESSION at the top . Fig 4 shows logout.php destroying the session and redirecting to login, confirming the user is fully signed out. Bonus features, TechHive also implements password reset incase a user forgets

their password, email verification on registration, two-factor authentication requiring a one-time code, and user profile management allowing customers to update their account details.

Week 8: Responsive Web Design and Mobile-First Development

Fig 1: Responsive Personal Profile Page

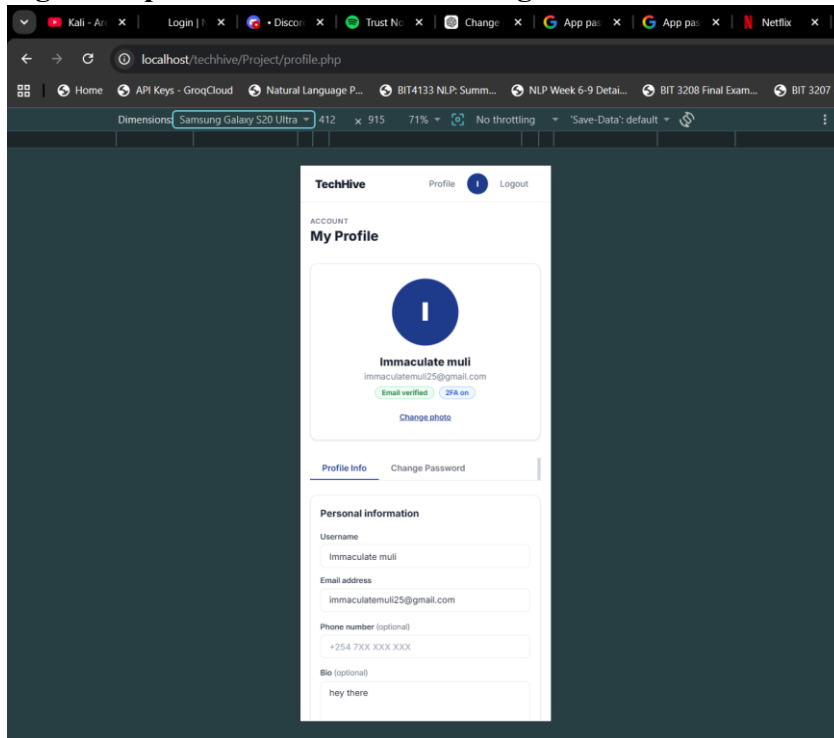


Fig 2: Responsive Product Showcase

Fig 2.1 Mobile view display

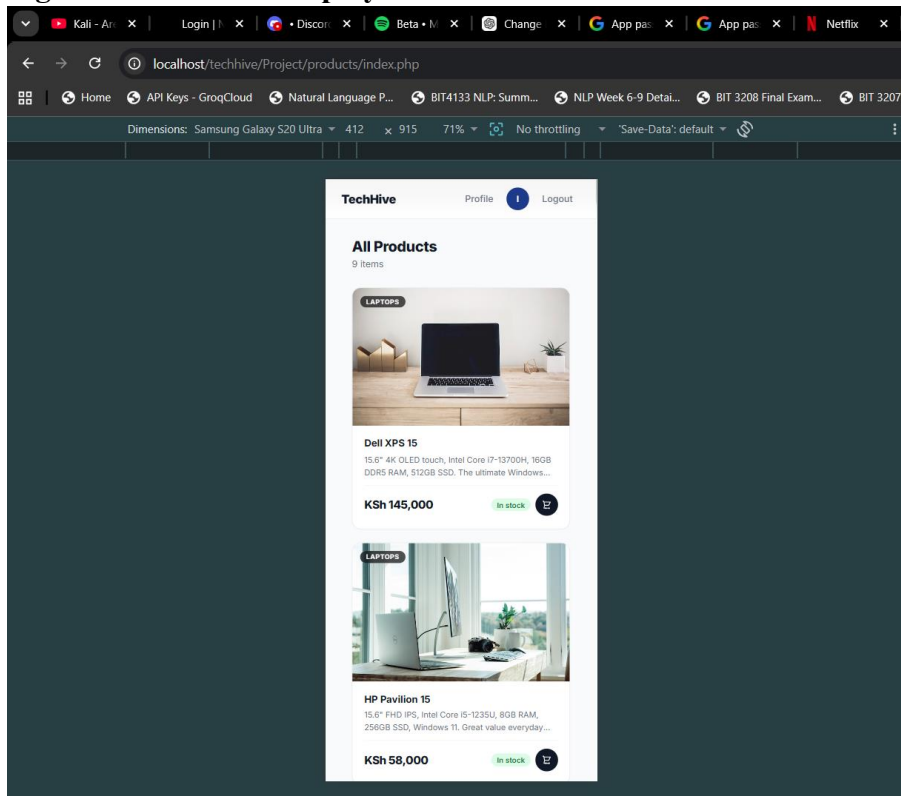


Fig 2.2 Tablet view display

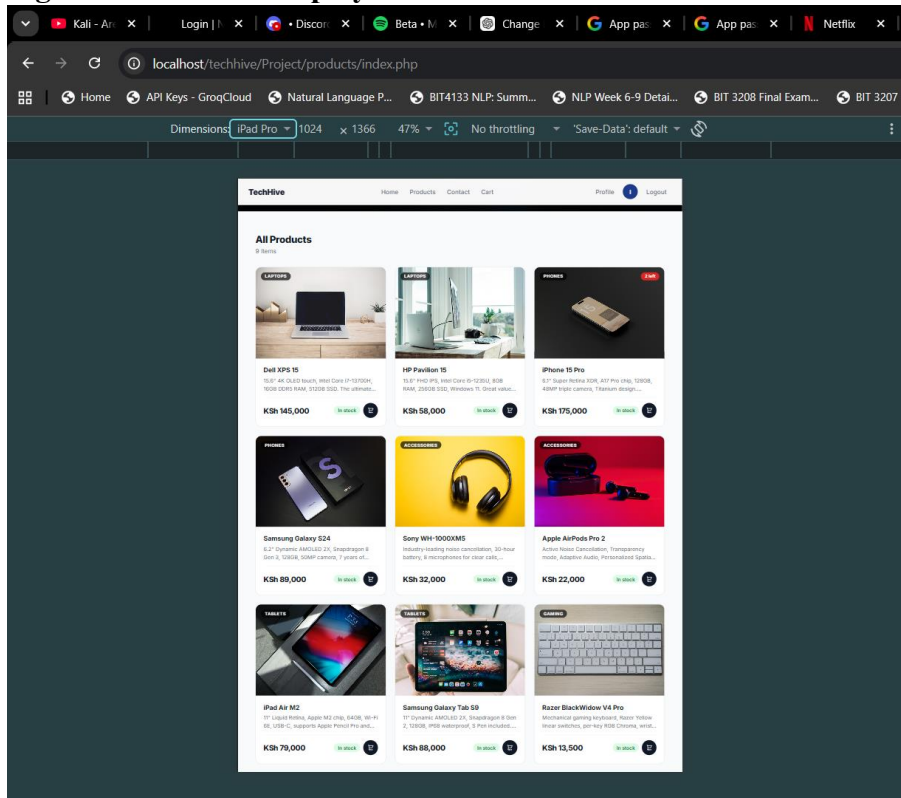
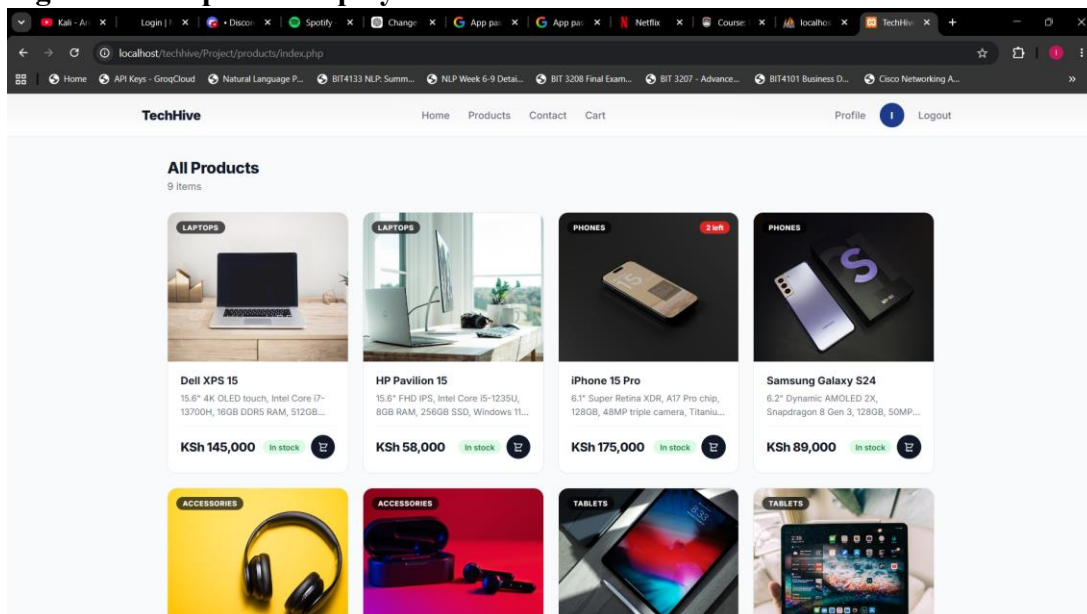


Fig 2.3 Desktop view display



Week 8 focused on Responsive Web Design and Mobile-First Development. Fig 1 shows the responsive personal profile page. Fig 2.1 shows the mobile view displaying one product per row, Fig 2.2 shows the tablet view displaying two products per row and Fig 2.3 shows the desktop view displaying four products per row .